

# A New Lower Bound for the Supremum of Autoconvolutions

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## Abstract

We establish a lower bound on the autoconvolution constant  $C_{1a} = \inf_f \|f * f\|_{L^\infty} / (\int f)^2$ , taken over nonnegative  $f \in L^1(\mathbb{R})$  supported on  $(-1/4, 1/4)$ :

$$C_{1a} \geq 1292/1000 = 1.292.$$

This improves on the published lower bound of 1.28 due to Cloninger and Steinerberger [CS17]<sup>1</sup> and on the rigorous analytic bound of 1.27481 established by Matolcsi and Vinuesa [MV10]. The argument extends the Matolcsi–Vinuesa master inequality: the single arcsine kernel is replaced by a convex combination of three arcsine kernels, and the trigonometric multiplier  $G$  is re-optimized as a 200-cosine expansion. An internal arb-coupled enclosure yields the slightly sharper  $M_{\text{cert}} \geq 66167/51200 \approx 1.29232$ , recorded for completeness but not encoded in the headline Lean theorem (see Section 5).

The five resulting functionals are evaluated rigorously in `flint.arb` interval arithmetic at 256-bit precision. The no- $z_1$  quadratic master inequality, applied to the rounded rational anchors, yields strict failure at  $M = 1292/1000$  with margin  $\tau - \Phi(M) \geq 9.6 \times 10^{-5}$ . The analytic reduction is formalized in Lean 4 across 30 modules ( $\approx 15.5$  thousand lines) rooted at `lean/Sidon/MultiScale.lean`, with two machine-checked headline theorems. A *conditional* headline `autoconvolution_ratio_ge_1292_1000` takes an `ExtremiserPrimitives`  $f$  hypothesis record encoding the outputs of MO 2009 Lemmas 3.1–3.4 / MV 2010 Lemma 3.1 at  $(f, K_{\text{ms}})$  — which the paper discharges by direct citation, exactly as MV 2010 cited MO 2009 for the single-arcsine instance — and reaches Lean’s three core axioms together with exactly two numerical user axioms recording the certifier’s outputs at the rational slacks (the  $L^2$  integral of  $K_{\text{ms}}$  and the gain ratio  $(4/u)m_G^2/S_1$ ). A second, *unconditional* headline `C1a_ge_1292_unconditional` strengthens this: it takes only the admissibility hypotheses on  $f$  (nonnegativity, support in  $(-1/4, 1/4)$ ,  $L^1 \cap L^2$ , unit mass) and constructs the bundle internally via `ExtremiserPrimitives.of_admissible`, so no hypothesis record is carried. Its dependency closure reaches the same three core axioms together with four numerical user axioms (the two above, plus the certified lower bound  $m_G \geq 998/1000$  and the active-set lattice positivity  $\widehat{K}_{\text{ms}}(j) > 0$  for  $1 \leq j \leq 200$ ). The quadratic inversion and the slack-soundness statements are Lean theorems throughout; see Section 5.3.

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<sup>1</sup>An unpublished 1.2802 figure is attributed in Tao’s `optimizationproblems` repository to Xie (2026, AI-assisted, unaudited) [Xie26]; it is not part of the published CS17 result.

# 1. Introduction

## 1.1. Overview

This subsection is an informal account of what is proved and why it matters; the precise definitions, the numerical record, and the formal statement of the result follow in Sections 1.2–1.4.

**The question.** Fix a nonnegative function  $f$  on the real line, supported in the short interval  $(-1/4, 1/4)$  and normalized so that  $\int f = 1$ ; think of  $f$  as a probability density. If  $X$  and  $Y$  are independent random variables each with density  $f$ , then their sum  $X + Y$  has density the autoconvolution  $f * f$ , itself a probability density, now supported in  $(-1/2, 1/2)$ . Since  $f * f$  integrates to 1 over an interval of length 1, its peak value  $\|f * f\|_{L^\infty}$  is at least 1, with equality only if  $f * f$  were the uniform density on  $(-1/2, 1/2)$  — a shape no admissible  $f$  produces. The *autoconvolution constant*

$$C_{1a} = \inf_f \|f * f\|_{L^\infty} / \left(\int f\right)^2$$

measures *how flat* an autoconvolution can be made: it is the smallest peak attainable as  $f$  ranges over all admissible shapes. Its exact value is unknown; the problem is to trap it between explicit numerical bounds, and in particular to push the lower bound upward.

**Why it matters.** This continuous extremal problem is the analytic shadow of a classical question in additive combinatorics going back to Erdős and Turán [ET41]: how large can a set of integers be if no integer has too many representations as a sum of two of its elements (a Sidon, or  $B_2$ , condition)? Martin and O’Bryant [MO09] made the link precise, showing that the asymptotics of such extremal sets are controlled by this same constant  $C_{1a}$ . Each improvement to the *rigorous* lower bound on  $C_{1a}$  sharpens the corresponding asymptotic results for  $B_2^+[1]$  and related Sidon variants, which is why the constant has drawn sustained attention for several decades.

**Why lower bounds are hard.** Exhibiting one function  $f$  can only ever yield an *upper* bound on the infimum  $C_{1a}$ . A *lower* bound is a statement of a different kind: it must certify, simultaneously and for *every* admissible  $f$ , that the peak cannot be small. Matolcsi and Vinuesa [MV10] supply exactly such a certificate. From an auxiliary pair chosen by us — a nonnegative “kernel”  $K$  whose Fourier coefficients are also nonnegative (a *Bochner-admissible* kernel) together with a trigonometric “multiplier”  $G$  — their duality produces a single quadratic inequality that every admissible  $f$  must satisfy, and hence a rigorous lower bound on  $C_{1a}$ . Any valid pair  $(K, G)$  yields *some* bound; the entire game is to construct a pair that yields a *good* one. With a single arcsine kernel and a 119-term cosine multiplier, Matolcsi and Vinuesa obtained  $C_{1a} \geq 1.27481$ ; on the other side, the constant is known to be at most  $\approx 1.502862$  [GGTW26].

**What is new here.** We enlarge the class of certificates. Matolcsi and Vinuesa worked with a kernel at a *single* scale. We use that a convex combination of arcsine kernels at *several* scales is still Bochner-admissible (Theorem 3.2), and that mixing scales removes a structural bottleneck. A single-scale arcsine kernel has “blind spots”: frequencies at which its Fourier transform nearly vanishes, namely the zeros of a Bessel function. Each near-zero sits in a denominator of the certificate and inflates a critical quantity  $S_1$ , throttling the resulting bound. Different scales place their blind spots at different frequencies, so a three-scale blend keeps the Fourier transform uniformly away from zero:  $S_1$  falls from  $\approx 87.4$  to  $\leq 29.841$ , and the bound rises in proportion. Capturing the gain also requires re-optimizing the multiplier  $G$  for the blended kernel, which we do

by convex programming with 200 cosine terms. The resulting three-scale construction — arcsine half-widths  $(138, 55, 25)/1000$  with weights  $(85, 10, 5)/100$  — gives the new lower bound

$$C_{1a} \geq 1292/1000 = 1.292,$$

(Theorem 1.1).

**In what sense this is rigorous.** Closing the certificate reduces to controlling five real numbers extracted from  $(K, G)$ . These are not merely estimated in floating point: each is enclosed in a mathematically guaranteed interval using `flint.arb` arbitrary-precision interval arithmetic at 256-bit precision [Joh17], and each interval is then rounded *outward* to an exact rational. The final quadratic step that produces 1.292 is carried out in exact rational arithmetic, so the conclusion is independent of any floating-point rounding. The trust base for the numerical step is therefore Lean’s logical core together with the soundness of `flint.arb`’s interval arithmetic [Joh17], a standard assumption in computer-assisted real-number results (cf. Flyspeck for Kepler).

Independently, the entire analytic reduction — the master inequality, its inversion, and the admissibility of the three-scale kernel — is mechanized in the Lean 4 proof assistant [dMU21, The20] (30 modules,  $\approx 15.5$  thousand lines, no `sorry`). The mechanization provides two machine-checked headline theorems. The *conditional* headline rests only on Lean’s three logical axioms together with exactly two explicitly identified numerical inputs: decidable inequalities about concrete real analytic expressions inside Lean — the  $L^2$  integral of the three-scale kernel  $K_{\text{ms}}$ , and the gain ratio  $(4/u) m_G^2/S_1$  formed from the embedded 200 rational QP coefficients and the closed-form Bessel-based periodic Fourier coefficients of  $K_{\text{ms}}$  — that play the role of the “Mathematica computed this value” citations in [MV10], here backed by certified interval arithmetic rather than heuristic numerics.

The standard Fourier-analytic facts of MO 2009 Lemmas 3.1–3.4 / MV 2010 Lemma 3.1 are applied directly to the admissible kernel  $K_{\text{ms}}$  (a pdf supported in  $[-\delta_1, \delta_1]$  with nonnegative periodic Fourier coefficients and  $K_{\text{ms}} \in L^2$ ); the paper discharges those outputs at  $(f, K_{\text{ms}})$  by direct citation, exactly as MV 2010 cited MO 2009 for the single-arcsine application. The conditional headline carries those facts as the named fields of an `ExtremiserPrimitives f` record. The mechanization then goes one step further than the paper-level citation: a dedicated `Sidon.Constructor` layer (17 axiom-free modules) constructs this record for *any* admissible  $f$  — mechanizing the  $L^1$  convolution Fourier identity,  $K_{\text{ms}} \in L^2$  via the Young estimate  $L^{4/3} \star L^{4/3} \rightarrow L^2$ , a period- $u$  Poisson sampling lemma (generalizing `mathlib`’s period-1/continuous version), MO 2009 Lemma 2.1 in period- $u$  Parseval form, the period-1 Parseval identity for  $f \circ f$ , and the Cauchy–Schwarz floor. The resulting *unconditional* headline `C1a_ge_1292_unconditional` takes only the admissibility hypotheses on  $f$  and discharges the bundle internally via `ExtremiserPrimitives.of_admissible`, removing the hypothesis record entirely (Section 5.3). It thereby establishes  $C_{1a} \geq 1.292$  to at least the rigor of Matolcsi–Vinuesa [MV10]: the analytic content is both cited (as MV did) and now mechanized in Lean, the assembly is carried out in exact rational arithmetic, and the only inputs taken on trust are four computer-checked numerical facts of the same *kind* as MV’s Mathematica citations, but verified in `flint.arb` interval arithmetic at 256-bit precision (rigorous, not heuristic) and corroborated independently in `mpmath`.

**Roadmap.** Section 2 recalls the Matolcsi–Vinuesa master inequality and its inversion in the form we need. Section 3 introduces the three-scale arcsine kernel and proves its admissibility. Section 4 certifies the five governing functionals as exact rational bounds via interval arithmetic.

Section 5 closes the inequality at the rational witness  $M = 1292/1000$ , proving the main theorem, and summarizes the Lean formalization.

## 1.2. The Constant $C_{1a}$

Let  $\mathcal{F} = \{f \in L^1(\mathbb{R}) : f \geq 0, \text{supp}(f) \subseteq (-1/4, 1/4), \int f > 0\}$ . (By the Schinzel–Schmidt reduction [SS02], we may further assume  $f \in L^2(\mathbb{R})$  without loss of generality when controlling the ratio  $R(f)$  from below; the period-1 Plancherel identity  $\sum_{j \in \mathbb{Z}} |\hat{f}(j)|^4 = \|f * f\|_{L^2}^2$  used in the master inequality’s tail bound is consequently well-defined.) The *autoconvolution constant* is

$$C_{1a} = \inf_{f \in \mathcal{F}} R(f), \quad R(f) := \frac{\|f * f\|_{L^\infty}}{(\int f)^2}.$$

By homogeneity we may normalize  $\int f = 1$ , in which case  $R(f) = \|f * f\|_{L^\infty} \geq 1$ . Heuristically,  $R(f)$  measures the peak height of the self-convolution  $f * f$  relative to its  $L^1$  mass, so  $C_{1a}$  identifies the flattest autoconvolution achievable in the class  $\mathcal{F}$ . This constant is the continuous analogue of the Sidon-set extremal problem [ET41, MO09]; improving its lower bound tightens known asymptotic results on  $B_2^+[1]$  sets and related Sidon variants.

## 1.3. Known Bounds

Prior lower bounds on  $C_{1a}$  are summarized in Table 1. Matolcsi–Vinuesa [MV10] established the rigorous analytic value 1.27481 in 2010, and Cloninger and Steinerberger [CS17] announced a computational improvement to 1.28 via a discrete branch-and-prune search. An unpublished improvement to 1.2802 is attributed in Tao’s `optimizationproblems` repository [Xie26] to Xie (2026, AI-assisted, unaudited). All subsequent published progress on the constant has been on the upper-bound side, where a recent AI-assisted search lowered the ceiling to 1.502862 [GGTW26]. The present work raises the lower bound to 1.292 (Theorem 1.1) by refining the kernel class entering the Matolcsi–Vinuesa dual framework.

| Bound       | Value           | Reference                                         |
|-------------|-----------------|---------------------------------------------------|
| Upper bound | $\leq 1.502862$ | Georgiev et al. [GGTW26] (AlphaEvolve)            |
| Upper bound | $\leq 1.50972$  | Matolcsi & Vinuesa [MV10]                         |
| Lower bound | $\geq 1.262$    | Martin & O’Bryant [MO09]                          |
| Lower bound | $\geq 1.27481$  | Matolcsi & Vinuesa [MV10]                         |
| Lower bound | $\geq 1.28$     | Cloninger & Steinerberger [CS17]                  |
| Lower bound | $\geq 1.2802$   | Xie [Xie26] (unpublished, AI-assisted, unaudited) |
| Lower bound | $\geq 1.292$    | <b>This work</b>                                  |

**Table 1:** Published bounds on  $C_{1a}$  in the normalization used here.

## 1.4. Strategy and Main Result

The Matolcsi–Vinuesa [MV10] dual framework (Section 2) parametrizes a Bochner-admissible kernel  $K$  and a cosine multiplier  $G$ , producing a quadratic lower bound on  $R(f)$  in five real functionals  $(k_1, K_2, S_1, m_G, a)$ , of which  $(k_1, K_2)$  are extracted from the kernel and  $(S_1, m_G, a)$  from the multiplier (with  $a = (4/u) m_G^2 / S_1$  a derived gain functional). MV 2010 takes  $K$  to be the arcsine kernel at half-width  $\delta = 138/1000$  and  $G$  a 119-cosine expansion, achieving  $R(f) \geq 1.27481$ . The present construction (i) replaces the single arcsine by a three-scale convex combination  $K_{\text{ms}}$  at half-widths  $(138, 55, 25)/1000$  with weights  $(85, 10, 5)/100$ ; (ii) re-optimizes  $G$  as a 200-cosine

expansion against  $K_{\text{ms}}$ ; (iii) certifies the five functionals in `flint.arb` interval arithmetic; and (iv) solves the `no- $z_1$`  quadratic master inequality with the certified rationals. The quadratic inequality  $\Phi(M) \geq \tau$  fails strictly at the rational witness  $M = 1292/1000$ , with margin  $9.6 \times 10^{-5}$  from the rational anchors, giving the main result.

**Theorem 1.1** (Main result). *Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be nonnegative with  $\text{supp}(f) \subseteq (-1/4, 1/4)$ ,  $\int f > 0$ , and  $\|f * f\|_{L^\infty} < \infty$ . Then*

$$\frac{\|f * f\|_{L^\infty}}{(\int f)^2} \geq \frac{1292}{1000}.$$

*In particular  $C_{1a} \geq 1292/1000 = 1.292$ .*

The proof is assembled in Section 5; all numerical anchors are reproducible from the accompanying certificate, and a Lean 4 mechanization is summarized at the end of Section 5.

## 2. The Matolcsi–Vinuesa Master Inequality

We summarize the dual framework of [MV10, §3]; proofs of the four ingredient lemmas and of the resulting quadratic are recorded in that reference. Throughout,  $f \in \mathcal{F}$  with  $\int f = 1$ ,  $\widehat{h}(\xi) = \int h(x)e^{-2\pi i \xi x} dx$ , and  $M = R(f) = \|f * f\|_{L^\infty}$ . The framework converts the desired lower bound on  $M$  into a quadratic inequality whose coefficients depend on two auxiliary objects of our choice, a positive-definite kernel  $K$  and a trigonometric multiplier  $G$ , so that any rigorous numerical control over  $(K, G)$  translates into a rigorous lower bound on  $M$ .

### 2.1. Admissible Kernels and Multipliers

**Definition 2.1** (Bochner-admissible kernel). Fix  $\delta \in (0, 1/4]$  and set  $u = 1/2 + \delta$ . A function  $K : \mathbb{R} \rightarrow \mathbb{R}$  is *Bochner-admissible at scale  $\delta$*  if

- (K1)  $K \geq 0$  a.e. on  $\mathbb{R}$ ;
- (K2)  $K(-x) = K(x)$  for every  $x \in \mathbb{R}$ ;
- (K3)  $\text{supp}(K) \subseteq [-\delta, \delta]$  and  $\int K = 1$ ;
- (K4) the periodic Fourier coefficients  $\widetilde{K}(j) := \widehat{K}(j/u)$  satisfy  $\widetilde{K}(j) \geq 0$  for every  $j \in \mathbb{Z}$ .

We write  $K_2 := \|K\|_{L^2(\mathbb{R})}^2$ .

**Definition 2.2** (Admissible cosine multiplier). Fix  $u$  and  $N \in \mathbb{N}$ . A trigonometric polynomial  $G(x) = \sum_{j=1}^N a_j \cos(2\pi j x / u)$  with  $a_j \in \mathbb{R}$  is *admissible* if there is  $m_G > 0$  with  $G(x) \geq m_G$  for every  $x \in [0, 1/4]$ . Define

$$S_1 = \sum_{j=1}^N \frac{a_j^2}{\widetilde{K}(j)}, \quad a = \frac{4}{u} \cdot \frac{m_G^2}{S_1}. \quad (1)$$

### 2.2. The Master Inequality and Its Inversion

**Theorem 2.3** (Master inequality,  $z_1$ -free form). *Let  $K$  be Bochner-admissible at scale  $\delta$ ,  $u = 1/2 + \delta$ , and  $G$  admissible of degree  $N$  with constant  $m_G$ , defining  $S_1$  and  $a$  by (1). Then for every  $f \in \mathcal{F}$  with  $\int f > 0$ ,*

$$M + 1 + \sqrt{(M-1)(K_2-1)} \geq \frac{2}{u} + a. \quad (2)$$

*Proof.* Matolcsi–Vinuesa [MV10, Lemma 3.3, Eq. (10)] prove the sharper form

$$M + 1 + 2z_1^2 k_1 + \sqrt{(M-1-2z_1^4)(K_2-1-2k_1^2)} \geq \frac{2}{u} + a, \quad (3)$$

where  $k_1 = \widehat{K}(1)$  and  $z_1 = |\widehat{f}(1)|$  are period-1 Fourier coefficients (so that  $\widetilde{K}(j) = \widehat{K}(j/u)$  of Definition 2.1 is the period- $u$  companion). The  $z_1$ -free designation reflects that  $z_1$  is a Fourier coefficient of the (unknown) extremal  $f$ , so it cannot be controlled analytically; the Cauchy–Schwarz step below absorbs  $z_1$  (together with  $k_1$ ) into the kernel-side quantity  $K_2$ , leaving an inequality that depends only on quantities we can certify. Their proof uses the arcsine kernel only through Bochner-admissibility (Definition 2.1); the same derivation therefore applies to any admissible  $K$ , in particular to the convex combination  $K_{\text{ms}}$ . The radicand of (3) is nonnegative by the same Parseval-type bounds as in [MV10, §3]: since  $f$  is supported in  $[-1/4, 1/4]$ , period-1 Plancherel gives  $\sum_{j \in \mathbb{Z}} |\widehat{f}(j)|^4 = \|f * f\|_{L^2}^2 \leq \|f * f\|_{L^\infty} = M$ , so  $2z_1^4 \leq M - 1$  for  $f \in \mathcal{F}$  with  $\int f = 1$ ; similarly  $\sum_{j \in \mathbb{Z}} |\widehat{K}(j)|^2 = \|K\|_{L^2}^2 = K_2$  gives  $2k_1^2 \leq K_2 - 1$  for any Bochner-admissible  $K$  with  $\int K = 1$ . Inequality (2) then follows from (3) by the Cauchy–Schwarz estimate  $\sqrt{ac} + \sqrt{bd} \leq \sqrt{(a+b)(c+d)}$  applied to the nonnegative quadruple  $(a, b, c, d) = (2z_1^4, M - 1 - 2z_1^4, 2k_1^2, K_2 - 1 - 2k_1^2)$ .  $\square$

**Remark 2.4** (From MV’s  $0.5747/\delta$  to the abstract  $K_2$ , and the  $z_1$ -absorption). Two translations separate the form (3) we use from the literal statement of [MV10, Lemma 3.3, Eq. (10)], and we record both explicitly. First, MV 2010’s Eq. (10) is stated for the *single-arcsine* specialization: its radicand contains  $\sqrt{0.5747/\delta - 1 - 2k_1^2}$ , where  $0.5747/\delta$  is the Martin–O’Bryant Lemma 3.2 plug-in  $\|K_{\text{arc}}(\delta; \cdot)\|_{L^2}^2 = \delta^{-1} \int J_0(\pi\xi)^4 d\xi \leq 0.5747/\delta$  for the single arcsine kernel. We replace  $0.5747/\delta$  throughout by the abstract quantity  $K_2 = \|K\|_{L^2(\mathbb{R})}^2$ . This substitution is valid for any Bochner-admissible  $K$  because MV’s proof of Eq. (10) ([MV10, §3, proof of Lemma 3.3]) uses the kernel only through (i)  $K \in L^2$  with  $K_2 = \|K\|_{L^2}^2$ , (ii)  $K \geq 0$ , and (iii) the Bochner positivity  $\widetilde{K}(j) \geq 0$  — it never uses the arcsine shape beyond these three facts, and the value  $0.5747/\delta$  enters only as the numerical evaluation of  $\|K\|_{L^2}^2$  at the single-arcsine choice. All three facts hold for  $K_{\text{ms}}$  (Definition 2.1, established in Section 3), so the same proof yields (3) with  $K_2 = K_2(K_{\text{ms}})$ . Second, the passage from the sharp  $z_1, k_1$ -form (3) to the  $z_1$ -free form (2) is the present paper’s Cauchy–Schwarz step  $\sqrt{ac} + \sqrt{bd} \leq \sqrt{(a+b)(c+d)}$ , applied to the nonnegative quadruple  $(2z_1^4, M - 1 - 2z_1^4, 2k_1^2, K_2 - 1 - 2k_1^2)$  of the proof above; nonnegativity of its entries is exactly the period-1 Plancherel bounds  $2z_1^4 \leq M - 1$  (for  $f * f$ ) and  $2k_1^2 \leq K_2 - 1$  (for  $K_{\text{ms}}$ ). In short: the cite to MV Eq. (10) supplies the underlying  $z_1, k_1$  structure; the  $0.5747/\delta \rightarrow K_2$  substitution and the  $z_1$ -absorption are this paper’s translations, each justified above.

**Lemma 2.5** (Quadratic inversion). *For fixed  $K_2 \geq 1$ , the function*

$$\Phi(M) := M + 1 + \sqrt{(M - 1)(K_2 - 1)}$$

*is continuous and strictly increasing on  $[1, \infty)$  with  $\Phi(1) = 2$ . For any  $\tau \geq 2$  the equation  $\Phi(M) = \tau$  has a unique solution  $M_* \in [1, \infty)$ , and every  $f \in \mathcal{F}$  with  $\Phi(R(f)) \geq \tau$  satisfies  $R(f) \geq M_*$ . In particular, with  $\tau = 2/u + a$ ,  $C_{1a} \geq M_*$ .*

*Proof.* For  $M > 1$ ,  $\Phi'(M) = 1 + \sqrt{(K_2 - 1)/(M - 1)}/2 > 1$ , so  $\Phi$  is strictly increasing on  $(1, \infty)$  and continuous at  $M = 1$  from the right. Existence, uniqueness, and the conclusion follow from the intermediate value theorem and Theorem 2.3.  $\square$

The MV 2010 single-arcsine optimum yields  $M_* \approx 1.2748$ . The remainder of the paper raises  $M_*$  by enlarging the kernel class.



### 2.3. Extension to Convex Combinations

Matolcsi–Vinuesa [MV10, Lemma 3.1] (which itself rests on Martin–O’Bryant [MO09, Lemmas 3.1–3.4]) was stated for a single arcsine kernel, but its proof uses  $K$  only through the Bochner-admissibility hypotheses (K1)–(K4) of Definition 2.1. We record the elementary extension to convex combinations here, since the kernel  $K_{\text{ms}}$  of Section 3 is constructed by precisely such a combination. Throughout,  $(f \circ f)(x) := \int_{\mathbb{R}} f(t) f(x+t) dt$  denotes the convolutional autocorrelation in MV’s notation [MV10, §3].

**Lemma 2.6** (MV Lemma 3.1 extension to convex-combination kernels). *Let  $K = \sum_{i=1}^n \lambda_i K_i$  be a finite convex combination of kernels  $K_i$ , each Bochner-admissible at some scale  $\delta_i \in (0, 1/4]$  in the sense of Definition 2.1, with  $\lambda_i \geq 0$  and  $\sum_i \lambda_i = 1$ . Then  $K$  is also Bochner-admissible at scale  $\delta_{\max} := \max_i \delta_i$ , and for every  $f \in \mathcal{F}$  with  $\int f = 1$  the four atomic identities of [MV10, Lemma 3.1] / [MO09, Lemmas 3.1–3.4], together with the  $z_1$ -absorption Cauchy–Schwarz step in the proof of Theorem 2.3, deliver the following bounds at the pair  $(f, K)$ :*

- (1)  $\int_{\mathbb{R}} (f * f)(x) K(x) dx \leq \|f * f\|_{L^\infty}.$
- (2)  $\int_{\mathbb{R}} (f \circ f)(x) K(x) dx \leq 1 + \sqrt{\|f * f\|_{L^\infty} - 1} \sqrt{K_2 - 1}.$
- (3)  $\frac{2}{u} + 2u \sum_{j \neq 0} (\operatorname{Re} \tilde{f}(j))^2 \tilde{K}(j) \leq \int_{\mathbb{R}} ((f * f)(x) + (f \circ f)(x)) K(x) dx$ , where  $u = 1/2 + \delta_{\max}$ ,  
 $\tilde{f}(j) := \frac{1}{u} \int_{-u/2}^{u/2} f(x) e^{-2\pi i j x / u} dx$  is the (period- $u$  averaged) Fourier coefficient of  $f$ , and  
 $\tilde{K}(j) := \tilde{K}(j/u)$  as in Definition 2.1 (see Remark 2.7).
- (4) For every even, real-valued,  $u$ -periodic trigonometric polynomial  $G$  with  $\tilde{G}(0) = 0$  and  $\min_{x \in [0, 1/4]} G(x) \geq m_G > 0$ ,

$$u^2 \sum_{j \neq 0} (\operatorname{Re} \tilde{f}(j))^2 \tilde{K}(j) \geq \frac{m_G^2}{\sum_{j: \tilde{G}(j) \neq 0} \tilde{G}(j)^2 / \tilde{K}(j)}.$$

*Proof. Preservation of (K1)–(K4) under convex combination.* Nonnegativity, evenness, and the support inclusion  $\operatorname{supp} K \subseteq \bigcup_i \operatorname{supp} K_i \subseteq [-\delta_{\max}, \delta_{\max}]$  are immediate. Unit mass and Fourier positivity follow from linearity:  $\int K = \sum_i \lambda_i \int K_i = \sum_i \lambda_i = 1$ , and  $\tilde{K}(j) = \tilde{K}(j/u) = \sum_i \lambda_i \tilde{K}_i(j) \geq 0$  since each  $\tilde{K}_i(j) \geq 0$  and  $\lambda_i \geq 0$ . Hence  $K$  satisfies (K1)–(K4) at scale  $\delta_{\max}$ .

*Identities (1)–(4) transport.* Each identity uses  $K$  only through (K1)–(K4):

- (1) is the pointwise bound  $(f * f) \cdot K \leq \|f * f\|_{L^\infty} \cdot K$  integrated using  $K \geq 0$  and  $\int K = 1$  (K1, K3).
- (2) is MV 2010 Lemma 3.3 / MO 2009 Lemma 3.2 (period-1 Parseval applied to  $f \circ f$  paired against  $K$ , followed by a sequence Cauchy–Schwarz, in their sharp  $z_1, k_1$ -form Eq. (3)’s analogue for the  $f \circ f$  piece), followed by the  $z_1$ -absorption Cauchy–Schwarz step  $\sqrt{ac} + \sqrt{bd} \leq \sqrt{(a+b)(c+d)}$  of Theorem 2.3. Both steps use  $K$  only through  $\tilde{K}(j) \geq 0$  (K4) and the finiteness of the energy  $K_2 = \|K\|_2^2$  of Definition 2.1. For a convex combination of  $L^2$  kernels the latter follows from Minkowski’s inequality; for the specific three-scale  $K_{\text{ms}}$  it is verified on the Fourier side via Plancherel in Lemma 3.3. (We do *not* use  $K \in L_{\text{loc}}^\infty$ : the arcsine autoconvolution is unbounded at the origin.)

- (3) is the  $\geq$ -direction of the torus Parseval identity

$$\int_{\mathbb{R}} ((f * f) + (f \circ f)) K \, dx = \frac{2}{u} + 2u \sum_{j \neq 0} (\operatorname{Re} \tilde{f}(j))^2 \tilde{K}(j),$$

which combines the torus Parseval split for  $(f * f) + (f \circ f)$  against  $K$ , the constant term  $2u (\operatorname{Re} \tilde{f}(0))^2 \tilde{K}(0) = 2/u$  (from  $\tilde{f}(0) = \frac{1}{u} \int f = \frac{1}{u}$  and  $\tilde{K}(0) = \widehat{K}(0) = \int K = 1$  by (K3)), and the Fourier expansion of the tail. We retain only the inequality direction; see Remark 2.7 for why the  $\geq$ -direction alone suffices and follows from  $\tilde{K}(j) \geq 0$  (K4) without the full period- $u$  summation.

- (4) is the discrete weighted Cauchy–Schwarz (Sedrakyan / Titu / Engel form)  $(\sum a_j b_j)^2 \leq (\sum a_j^2 \tilde{K}(j)) (\sum b_j^2 / \tilde{K}(j))$  applied with  $a_j = \operatorname{Re} \tilde{f}(j) \sqrt{\tilde{K}(j)}$  and  $b_j = \tilde{G}(j) / \sqrt{\tilde{K}(j)}$  (well-defined for  $j$  with  $\tilde{K}(j) > 0$ ), combined with the inner-product floor

$$u \cdot \sum_{j \neq 0} \operatorname{Re} \tilde{f}(j) \tilde{G}(j) \geq m_G, \quad (4)$$

established as follows. By admissibility  $f \geq 0$  with  $\operatorname{supp} f \subseteq (-1/4, 1/4)$ , and  $\int_{\mathbb{R}} f \, dx = 1$  under the normalization  $\int f = 1$  (Section 1.2). Since  $f$  is real-valued and even,  $\tilde{f}(j) \in \mathbb{R}$ , and the period- $u$  Parseval pairing of the  $u$ -periodic, real, even multiplier  $G$  against  $f$  reads

$$\int_{\mathbb{R}} f(x) G(x) \, dx = u \sum_{j \in \mathbb{Z}} \operatorname{Re} \tilde{f}(j) \tilde{G}(j)$$

(MO 2009 Lemma 2.1, applied to the pairing of  $f$ , supported in  $(-1/4, 1/4)$ , against  $G$ ). The  $j = 0$  term is  $u \operatorname{Re} \tilde{f}(0) \tilde{G}(0) = (\int f) \tilde{G}(0)$  (using  $\tilde{f}(0) = \frac{1}{u} \int f$ ), which vanishes because the cosine multiplier  $G(x) = \sum_{j=1}^N a_j \cos(2\pi j x / u)$  of Definition 2.2 has no constant term, i.e.  $\tilde{G}(0) = 0$ . Finally  $G(x) \geq m_G$  for  $x \in [-1/4, 1/4] \supseteq \operatorname{supp} f$  together with  $f \geq 0$  gives the pointwise bound  $m_G f(x) \leq f(x) G(x)$  for all  $x$  (trivially for  $x \notin \operatorname{supp} f$ ), so that

$$u \sum_{j \neq 0} \operatorname{Re} \tilde{f}(j) \tilde{G}(j) = \int_{\mathbb{R}} f(x) G(x) \, dx \geq m_G \int_{\mathbb{R}} f(x) \, dx = m_G,$$

which is (4). (The Lean witness is `Sidon.MVLemmas.mv_inner_product_floor`, which integrates the pointwise bound  $m_G f \leq f G$  and substitutes the torus-Parseval identity.)

Each step invokes (K1)–(K4) alone; no specific kernel structure is used. Hence the identities transport unchanged to any  $K = \sum_i \lambda_i K_i$  satisfying (K1)–(K4). The formal Lean witness of this extension is the module `Sidon.MVLemmas`, where identities (1)–(4) are proven for a *generic* admissible  $K$  (with the four atomic primitives of (2) and (3) carried as named hypotheses, exactly mirroring the Fourier-analytic ingredients listed above).  $\square$

**Remark 2.7** (Why item (3) is invoked only in its  $\geq$ -direction). MO 2009 Lemma 3.3 / MV 2010 Lemma 3.3 state item (3) as an *equality*,

$$\int_{\mathbb{R}} ((f * f) + (f \circ f)) K \, dx = \frac{2}{u} + 2u \sum_{j \neq 0} (\operatorname{Re} \tilde{f}(j))^2 \tilde{K}(j),$$

and the master inequality of Theorem 2.3 chains against this only through the lower bound on the left-hand side, i.e. the  $\geq$ -direction printed in Lemma 2.6(3). The equality of course implies the



inequality by reflexivity, so the citation to MO 2009 / MV 2010 remains valid; we record here that the  $\geq$ -direction needs strictly less. Indeed, since  $\widetilde{K}(j) \geq 0$  for every  $j$  (K4) and  $(\operatorname{Re} \widetilde{f}(j))^2 \geq 0$ , every term of the tail sum is nonnegative, so for any finite  $J \subseteq \mathbb{Z} \setminus \{0\}$  the truncated identity

$$\int_{\mathbb{R}} ((f * f) + (f \circ f)) K \, dx \geq \frac{2}{u} + 2u \sum_{j \in J} (\operatorname{Re} \widetilde{f}(j))^2 \widetilde{K}(j)$$

follows from finite- $J$  Parseval and Bochner positivity alone, with the full sum recovered by monotone convergence; no period- $u$  Poisson summation over all of  $\mathbb{Z}$  is required. This is precisely the refactor that lets the Lean formalization (`Sidon.MultiScale`, field `hEq3_ge`; supported by `Sidon.MVLemmas.mv_eq3_ge`) discharge item (3) from `mathlib`'s  $L^1 \cap L^2$  Plancherel and finite- $J$  Parseval primitives, without a full period- $u$  Parseval API.

**Remark 2.8** (Boundary case of MO 2009 Lemma 2.1). The period- $u$  Parseval identity invoked in step (3) above is MO 2009 Lemma 2.1 applied to the *bilinear* pairing of  $(f * f) + (f \circ f)$  (supported in  $(-1/2, 1/2)$ ) against  $K = K_{\text{ms}}$  (supported in  $[-\delta_{\text{max}}, \delta_{\text{max}}]$ ). The hypothesis of MO Lemma 2.1 requires  $\alpha_1 + \alpha_2 \leq u$  where  $\alpha_1, \alpha_2$  are the support half-widths of the two functions in the pairing. Here  $\alpha_1 = 1/2$  and  $\alpha_2 = \delta_{\text{max}}$ , so the hypothesis reads  $1/2 + \delta_{\text{max}} \leq u$ , satisfied with equality by our choice  $u = 1/2 + \delta_{\text{max}}$  — the same boundary case used by MV 2010 at the single-arcsine point  $\delta = \delta_1 = 0.138$ . In particular, the hypothesis is NOT that each of  $f * f, f \circ f, K$  individually fit in one period  $(-u/2, u/2)$ ; only the bilinear sum-of-supports condition is needed.

We are explicit that MO 2009 Lemma 2.1 is *stated* with the weak inequality  $\alpha_1 + \alpha_2 \leq u$  and that we invoke it at the boundary equality  $\alpha_1 + \alpha_2 = u$ ; the justification of the equality case below is the present paper's analytic argument, not a verbatim invocation of MO. The argument is a passage to  $L^2$ -class representatives. Lemma 2.1 is a statement about  $L^2$  equivalence classes: its bilinear pairing is unchanged when either factor is modified on a Lebesgue-null set [Fol99, §2.1]. Now  $K_{\text{ms}} = \sum_i \lambda_i \eta_{\delta_i} * \eta_{\delta_i}$  is continuous near  $\pm\delta_{\text{max}}$  and vanishes there — near each endpoint the supports of the two arcsine factors  $\eta_{\delta_i} \subseteq (-\delta_i/2, \delta_i/2)$  meet only at a single point, so the convolution integral is zero — and in any case  $\{\pm\delta_{\text{max}}\}$  is Lebesgue-null, so  $K_{\text{ms}}$  represents the same  $L^2$  class as a function supported in the *open* interval  $(-\delta_{\text{max}}, \delta_{\text{max}})$ ; likewise  $f \in L^2$  may be taken supported in the open  $(-1/4, 1/4)$ . For these representatives  $\alpha_1 + \alpha_2 < u$  holds with strict inequality on the (open) support sets, and the pairing value is in any case insensitive to the measure-zero modification. This is consistent with how MO 2009 themselves treat the boundary: their proof of Lemma 3.3 reads “*Since the inequality  $\frac{1}{2} + \delta \leq u$  is satisfied, we may apply Lemma 2.1*” at exactly  $u = \frac{1}{2} + \delta$ , so the equality case is one already exercised in the source.

The period-1 Parseval identity used in step (2) is MO 2009 Lemma 2.2, applied to  $f \circ f$  and  $K$  both supported in  $(-1/2, 1/2)$  (the standard  $L^2$  Parseval on a single period).

The master inequality (3) of Theorem 2.3 is the algebraic combination (1)+(2)+(3)+(4) of these four identities together with the Parseval-type bounds  $2z_1^4 \leq \|f * f\|_{L^\infty} - 1$  and  $2k_1^2 \leq K_2 - 1$ . Lemma 2.6 therefore justifies the application of Theorem 2.3 to the three-scale kernel  $K_{\text{ms}}$  of Section 3.

### 3. The Three-Scale Arcsine Kernel

#### 3.1. Single-Scale Arcsine and the Sonine Identity

For  $\delta \in (0, 1/4]$ , let

$$\eta_\delta(x) := \frac{1}{\delta} \cdot \frac{2/\pi}{\sqrt{1 - (2x/\delta)^2}} \mathbf{1}_{|x| < \delta/2}$$

denote the arcsine density on  $(-\delta/2, \delta/2)$  (the pushforward of the arcsine density on  $(-1/2, 1/2)$  under  $u \mapsto \delta u$ ). By the Mehler–Sonine integral representation  $J_0(z) = (2/\pi) \int_0^1 \cos(zt)/\sqrt{1-t^2} dt$  [DLMF, §10.9.4],  $\widehat{\eta}_\delta(\xi) = J_0(\pi\delta\xi)$ . The *arcsine kernel at scale  $\delta$*  is the auto-convolution

$$K_{\text{arc}}(\delta; \cdot) := \eta_\delta * \eta_\delta,$$

which is nonnegative and even, supported on  $[-\delta, \delta]$ , with  $\int K_{\text{arc}}(\delta; \cdot) = (\int \eta_\delta)^2 = 1$ . The convolution theorem then gives

$$\widehat{K_{\text{arc}}(\delta; \cdot)}(\xi) = \widehat{\eta}_\delta(\xi)^2 = J_0(\pi\delta\xi)^2 \geq 0. \quad (5)$$

In particular each  $K_{\text{arc}}(\delta; \cdot)$  is Bochner-admissible at scale  $\delta$ , with periodic coefficients  $\widetilde{K_{\text{arc}}(\delta; j)} = J_0(\pi j \delta / u)^2$  for  $u \geq 2\delta$ .

### 3.2. The Convex Combination

**Definition 3.1** (Three-scale kernel). Fix

$$(\delta_1, \delta_2, \delta_3) = \left(\frac{138}{1000}, \frac{55}{1000}, \frac{25}{1000}\right), \quad (\lambda_1, \lambda_2, \lambda_3) = \left(\frac{85}{100}, \frac{10}{100}, \frac{5}{100}\right), \quad (6)$$

and set  $u = 1/2 + \delta_1 = 638/1000$ . The *three-scale arcsine kernel* is

$$K_{\text{ms}}(x) := \sum_{i=1}^3 \lambda_i K_{\text{arc}}(\delta_i; x), \quad x \in \mathbb{R}.$$

| $i$ | $\delta_i$ | $\lambda_i$ | $u$      | $N$ |
|-----|------------|-------------|----------|-----|
| 1   | 138/1000   | 85/100      |          |     |
| 2   | 55/1000    | 10/100      | 638/1000 | 200 |
| 3   | 25/1000    | 5/100       |          |     |

**Table 2:** Parameters of the three-scale arcsine kernel and the cosine multiplier. The period  $u$  and the cosine degree  $N$  are shared across the three scales.

### 3.3. Admissibility

**Theorem 3.2** (Admissibility). *The kernel  $K_{\text{ms}}$  is Bochner-admissible at scale  $\delta_1 = 138/1000 < 1/4$ , with periodic coefficients*

$$\widetilde{K_{\text{ms}}}(j) = \sum_{i=1}^3 \lambda_i J_0\left(\frac{\pi j \delta_i}{u}\right)^2 \geq 0, \quad j \in \mathbb{Z}. \quad (7)$$

*Proof.* We check conditions (K1)–(K4) of Definition 2.1.

- (K1) *Nonnegativity*, (K2) *evenness*. Each  $\lambda_i K_{\text{arc}}(\delta_i; \cdot) \geq 0$  and is even; nonnegative sums of even nonnegative functions are even and nonnegative.
- (K3) *Support and unit mass*.  $\text{supp } K_{\text{arc}}(\delta_i; \cdot) \subseteq [-\delta_i, \delta_i] \subseteq [-\delta_1, \delta_1]$  for  $i \geq 1$ , so  $\text{supp } K_{\text{ms}} \subseteq [-\delta_1, \delta_1]$ ; and  $\int K_{\text{ms}} = \sum_i \lambda_i \int K_{\text{arc}}(\delta_i) = \sum_i \lambda_i = 1$ .
- (K4) *Fourier positivity*. Linearity of the Fourier transform and (5) give

$$\widehat{K_{\text{ms}}}(\xi) = \sum_i \lambda_i J_0(\pi \delta_i \xi)^2 \geq 0,$$

hence (7). □

Two analytical properties of  $K_{\text{ms}}$  used silently in Sections 4–5 merit explicit statement. Both are immediate from the convex-combination structure and the corresponding single-scale facts; we record them as sub-lemmas to make the dependence on (K1)–(K4) of Definition 2.1 transparent.

**Lemma 3.3** ( $K_{\text{ms}} \in L^2(\mathbb{R})$ ).  $\|K_{\text{ms}}\|_{L^2(\mathbb{R})} \leq \sum_{i=1}^3 \lambda_i \|\eta_{\delta_i} * \eta_{\delta_i}\|_{L^2(\mathbb{R})} < \infty$ .

*Proof.* By Minkowski’s inequality on  $L^2(\mathbb{R})$  applied to the finite combination  $K_{\text{ms}} = \sum_i \lambda_i (\eta_{\delta_i} * \eta_{\delta_i})$ ,  $\|K_{\text{ms}}\|_2 \leq \sum_i \lambda_i \|\eta_{\delta_i} * \eta_{\delta_i}\|_2$ , so it suffices that each single-scale autoconvolution lies in  $L^2(\mathbb{R})$ . We argue on the Fourier side, where the singularity is invisible. By the Mehler–Sonine identity (5),  $\widehat{\eta_{\delta_i} * \eta_{\delta_i}}(\xi) = J_0(\pi \delta_i \xi)^2$ , so by Plancherel and the substitution  $t = \delta_i \xi$ ,

$$\|\eta_{\delta_i} * \eta_{\delta_i}\|_2^2 = \int_{\mathbb{R}} J_0(\pi \delta_i \xi)^4 d\xi = \frac{1}{\delta_i} \int_{\mathbb{R}} J_0(\pi t)^4 dt.$$

The Watson envelope  $|J_0(z)| \leq \sqrt{2/(\pi z)}$  for  $z \geq 1$  gives  $J_0(\pi t)^4 = O(t^{-2})$  as  $|t| \rightarrow \infty$ , while  $J_0(\pi t)^4 \leq 1$  on  $[-1, 1]$ ; hence the integral converges and each autoconvolution is in  $L^2(\mathbb{R})$ , so the right-hand side is finite. We emphasize that  $\eta_{\delta_i} * \eta_{\delta_i}$  is *not* bounded: it carries a logarithmic singularity at the *origin* (where  $(\eta_{\delta_i} * \eta_{\delta_i})(0) = \int \eta_{\delta_i}^2 = \infty$ ), not at the endpoints. Square-integrability is genuine— $\log^2$  is locally integrable—but is established here via Plancherel, not via any local boundedness of  $K_{\text{ms}}$ .  $\square$

**Lemma 3.4** (Bochner positivity of  $\widehat{K_{\text{ms}}}$ ). *For every  $\xi \in \mathbb{R}$ ,*

$$\widehat{K_{\text{ms}}}(\xi) = \sum_{i=1}^3 \lambda_i J_0(\pi \delta_i \xi)^2 \geq 0.$$

*In particular  $\widehat{K_{\text{ms}}}(j) \geq 0$  for every  $j \in \mathbb{Z}$ .*

*Proof.* By linearity of the Fourier transform,  $\widehat{K_{\text{ms}}} = \sum_i \lambda_i \widehat{\eta_{\delta_i} * \eta_{\delta_i}}$ . The convolution theorem and the Mehler–Sonine identity (5) give  $\widehat{\eta_{\delta_i} * \eta_{\delta_i}}(\xi) = \widehat{\eta_{\delta_i}}(\xi)^2 = J_0(\pi \delta_i \xi)^2$ . Each summand  $\lambda_i J_0(\pi \delta_i \xi)^2$  is nonnegative ( $\lambda_i \geq 0$ , squares are nonnegative), and the sum of nonnegatives is nonnegative. Setting  $\xi = j/u$  for  $j \in \mathbb{Z}$  yields the periodic statement.  $\square$

Lemma 3.3 discharges the finite- $K_2$  hypothesis underlying identity (2) of Lemma 2.6; Lemma 3.4 discharges (K4), which is the only kernel-dependent hypothesis used in (3) and (4).

The motivation for the multi-scale construction is structural: in (1),  $S_1$  blows up at frequencies  $j$  where the single-scale  $J_0(\pi j \delta / u)^2$  is close to a Bessel zero. Mixing three scales shifts the zeros of each summand, so the sum stays bounded away from zero on  $1 \leq j \leq 200$  (numerically  $\min_j \widehat{K_{\text{ms}}}(j) \geq 2.08 \times 10^{-4}$ ). This drops  $S_1$  from  $\approx 87.4$  (single-scale) to  $\leq 29.841$  (Lemma 4.3), and proportionally raises the gain  $a$ .

## 4. The Rigorous Numerical Certificate

### 4.1. The Five Certified Anchors

For the master inequality applied to  $(K_{\text{ms}}, G)$  we need rigorous bounds on five real functionals. We replace each by a rational *anchor*: a floor when the functional enters the master inequality on the side that weakens the bound under understatement (i.e. with a  $\geq$  sign), a ceiling on the side that weakens it under overstatement ( $\leq$ ), in both cases obtained by outward rounding of the certifier’s arb interval enclosure. Anchoring at exact rationals lets the final inversion (Section 5) be

carried out in exact rational arithmetic, independent of the floating-point pipeline that produced the enclosures. The cosine multiplier  $G(x) = \sum_{j=1}^N a_j \cos(2\pi jx/u)$  with  $N = 200$  has rational coefficients  $a_j \in \mathbb{Q}$  with denominator  $10^8$ , obtained by rounding the solution of the convex QP that minimizes  $\sum_j a_j^2 / \widehat{K}_{\text{ms}}(j)$  subject to  $G \geq 1$  on a uniform grid of 5001 points in  $[0, 1/4]$  (solved by MOSEK). The five lemmas below are arb-interval computations [Joh17]; all certified outputs are rounded outward to exact rationals. Only four of the five anchors ( $K_2$ ,  $S_1$ ,  $m_G$ ,  $a$ ) enter the  $z_1$ -free form (2) used in Section 5; the kernel-mass moment  $k_1$  is recorded because the certifier internally uses the sharper inequality (3) (which sharpens  $M_{\text{cert}}$  from 1.292 to  $\approx 1.29232$  before rounding to the rational headline target).

**Lemma 4.1** (Kernel-mass moment).  $k_1 := \widehat{K}_{\text{ms}}(1) = \sum_{i=1}^3 \lambda_i J_0(\pi \delta_i)^2 \geq 9212/10000$ .

*Proof.* At 256-bit precision, `arb.bessel_j(0)` evaluated at exact rational  $\pi \delta_i$  returns balls of radius  $< 7 \times 10^{-77}$ ; the certified sum is  $k_1 \geq 0.92124658$ , with slack  $\geq 4.6 \times 10^{-5}$  above 9212/10000.  $\square$

**Lemma 4.2** (Kernel-energy moment).  $K_2 = \|K_{\text{ms}}\|_{L^2(\mathbb{R})}^2 \leq 47897/10000$ .

*Proof.* Split  $K_2 = 2 \int_0^T \widehat{K}_{\text{ms}}(\xi)^2 d\xi + 2 \int_T^\infty \widehat{K}_{\text{ms}}(\xi)^2 d\xi$  with  $T = 10^5$ . The bulk integral is computed by arb adaptive Gauss–Legendre quadrature and certified at 4.78882342... with arb radius below  $10^{-10}$ . For the tail, the uniform bound  $|J_0(z)|^2 \leq 2/(\pi z)$  [Wat44] holds for  $z > 0$ ; since  $\xi \geq T = 10^5$  in the tail forces  $z = \pi \delta_i \xi \geq \pi(25/1000)(10^5) \approx 7854$ , the bound applies, and applied to both Bessel factors gives  $J_0(\pi \delta_i \xi)^2 J_0(\pi \delta_j \xi)^2 \leq 4/(\pi^4 \delta_i \delta_j \xi^2)$ , whence  $\widehat{K}_{\text{ms}}(\xi)^2 = \sum_{i,j} \lambda_i \lambda_j J_0(\pi \delta_i \xi)^2 J_0(\pi \delta_j \xi)^2 \leq 4C^2/(\pi^4 \xi^2)$  with  $C = \sum_i \lambda_i / \delta_i \approx 9.978$ . Integrating  $\int_T^\infty \xi^{-2} d\xi = 1/T$  and doubling for both half-lines,  $2 \int_T^\infty \widehat{K}_{\text{ms}}^2 \leq 8C^2/(\pi^4 T) \leq 8.19 \times 10^{-5}$ . Summing,  $K_2 \in [4.78882342, 4.78890519] \subseteq [0, 47897/10000]$ , slack  $\geq 7.9 \times 10^{-4}$ .  $\square$

**Lemma 4.3** (Multiplier denominator).  $S_1 = \sum_{j=1}^{200} \frac{a_j^2}{\widehat{K}_{\text{ms}}(j)} \leq 29841/1000$ .

*Proof.* Each  $\widehat{K}_{\text{ms}}(j)$  is evaluated in arb from (7) with radius  $< 10^{-70}$ ; dividing exact rational squares  $a_j^2$  yields a certified sum  $S_1 \leq 29.840907$ , slack  $\geq 9.3 \times 10^{-5}$ .  $\square$

**Lemma 4.4** (Pointwise lower bound of  $G$ ).  $m_G := \min_{x \in [0, 1/4]} G(x) \geq 998/1000$ .

*Proof.* Partition  $[0, 1/4]$  into  $n = 32768$  closed cells of half-width  $r = 1/(8n) \approx 3.81 \times 10^{-6}$ . On each cell centred at  $c$ , the second-order Taylor expansion of  $G$  yields the rigorous arb enclosure

$$G(\text{cell}) \subseteq G(c) + G'(c) \cdot [-r, r] + G''(\text{cell}) \cdot [-r^2/2, r^2/2],$$

with  $G''$  evaluated as an arb interval over the entire cell from

$$G''(x) = - \sum_{j=1}^{200} a_j (2\pi j/u)^2 \cos(2\pi jx/u).$$

The minimum of the lower endpoints over all  $n$  cells gives the certified bound  $m_G \geq 0.99997987$ , slack  $\geq 1.9 \times 10^{-3}$  above 998/1000.  $\square$

**Lemma 4.5** (Gain functional).  $a = (4/u) \cdot m_G^2 / S_1 \geq 20925/100000$ .

*Proof.* Substituting the rational anchors of Lemmas 4.3 and 4.4 into the definition of  $a$  in (1),

$$a \geq \frac{4}{638/1000} \cdot \frac{(998/1000)^2}{29841/1000} = \frac{4000 \cdot 998^2}{638 \cdot 29841 \cdot 10^3} = \frac{3984016000}{19038558000} = 0.20926\dots,$$

hence  $a \geq 20925/100000$ , slack  $\geq 1.0 \times 10^{-5}$ . (The underlying arb interval evaluation returns the tighter enclosure  $a \geq 0.21009214$  when  $m_G$  and  $S_1$  are coupled rather than rounded separately; we use only the looser rational floor in what follows.)  $\square$

| Functional             | Symbol | Direction | Rational bound | Decimal |
|------------------------|--------|-----------|----------------|---------|
| Kernel-mass moment     | $k_1$  | $\geq$    | 9212/10000     | 0.9212  |
| Kernel-energy          | $K_2$  | $\leq$    | 47897/10000    | 4.7897  |
| Multiplier denominator | $S_1$  | $\leq$    | 29841/1000     | 29.841  |
| Multiplier minimum     | $m_G$  | $\geq$    | 998/1000       | 0.998   |
| Gain                   | $a$    | $\geq$    | 20925/100000   | 0.20925 |

**Table 3:** Certified rational anchors for the three-scale kernel  $K_{\text{ms}}$  at parameters (6) with  $u = 638/1000$  and the 200-cosine multiplier  $G$ .

## 4.2. Rescaling Invariance

Although the proof of Lemma 4.4 produces  $m_G \geq 998/1000$ , the master inequality is invariant under  $G \mapsto G/m_G$ : rescaling multiplies both  $S_1$  and  $m_G^2$  by  $1/m_G^2$ , leaving  $a$  unchanged. The certified rational bound  $m_G \geq 998/1000$  is therefore sufficient for the conclusion, and the lemma is not sharpened further.

## 4.3. Reproducibility

All five anchors are produced by the reproducible script and the associated certificate JSON: each entry is the outward-rounded enclosure returned by `flint.arb` at 256-bit precision; full pipeline inputs (the rational  $a_j$ , the bisection schedule, the quadrature tolerance) and the auditor logs are included in the accompanying repository.

# 5. Proof of the Main Theorem

## 5.1. The Strict-Failure Witness

We close the master inequality using the anchors of Table 3. The strategy is to exhibit a rational *witness*  $M_0 = 1292/1000$  for which  $\Phi(M_0) < \tau$  holds strictly; combined with the master inequality  $\Phi(R(f)) \geq \tau$ , this rules out  $R(f) \leq M_0$  by strict monotonicity of  $\Phi$  (Lemma 2.5), forcing  $R(f) > M_0$ . No analytic optimization over  $M$  is needed: the rational  $M_0$  is fixed in advance and the verification is purely arithmetic. Set  $\tau := 2/u + a$  and recall  $\Phi(M) := M + 1 + \sqrt{(M-1)(K_2-1)}$  from Lemma 2.5. With the certified rationals from Table 3,

$$\tau \geq \frac{2}{638/1000} + \frac{20925}{100000} = 3.134796\dots + 0.20925 = 3.344046\dots$$

**Proposition 5.1** (Strict failure at the rational witness). *With  $K_2 \leq 47897/10000$  and  $\tau = 2/u + a \geq 2000/638 + 20925/10^5$ ,*

$$\Phi(1292/1000) \leq 3.34395 < 3.344046\dots \leq \tau, \tag{8}$$

with margin  $\tau - \Phi(1292/1000) \geq 9.6 \times 10^{-5}$ .

*Proof.* At  $M = 1292/1000$ , using the certified upper bound  $K_2 \leq 47897/10000$ ,

$$(M-1)(K_2-1) \leq \frac{292}{1000} \cdot \frac{37897}{10000} = \frac{11065924}{10^7} = 1.1065924,$$

and since  $\Phi$  is increasing in  $K_2$ , this upper bound transfers to  $\Phi(M)$ . The rational bound  $(105195/10^5)^2 = 11065988025/10^{10} > 11065924/10^7$  gives  $\sqrt{(M-1)(K_2-1)} \leq 105195/10^5 = 1.05195$ , so

$$\Phi(1292/1000) \leq \frac{1292}{1000} + 1 + \frac{105195}{10^5} = \frac{66879}{20000} = 3.34395.$$

Combining with  $\tau \geq 4267003/1276000 = 3.344046 \dots$  yields

$$\tau - \Phi(1292/1000) \geq \frac{4267003}{1276000} - \frac{66879}{20000} = \frac{307}{3190000} \geq 9.6 \times 10^{-5} > 0.$$

The conclusion is independently re-certified by an arb cell-search bisection at 256-bit precision, which returns the sharper enclosure  $M_{\text{cert}} \geq 1.29232$  when the certifier's full arb interval bounds on all five functionals (rather than the rounded rational anchors of Table 3) are used.  $\square$

## 5.2. Assembly

**Theorem 5.2** (Restatement of Theorem 1.1). *Let  $f \in \mathcal{F}$  with  $\|f * f\|_{L^\infty} < \infty$ . Then  $R(f) \geq 1292/1000$ , and hence  $C_{1a} \geq 1292/1000$ .*

*Proof.* By homogeneity assume  $\int f = 1$ , so  $M := R(f) = \|f * f\|_{L^\infty} \geq 1$ . We assemble in five steps.

- (1) *Admissibility.* By Theorem 3.2,  $K_{\text{ms}}$  is Bochner-admissible at scale  $\delta_1 = 138/1000$ , with period  $u = 638/1000$ .
- (2) *Anchors.* By Lemmas 4.1–4.5, the multiplier  $G$  and the five rational bounds  $k_1 \geq 9212/10000$ ,  $K_2 \leq 47897/10000$ ,  $S_1 \leq 29841/1000$ ,  $m_G \geq 998/1000$ , and  $a \geq 20925/100000$  hold.
- (3) *Master inequality.* Theorem 2.3 applied to  $(K_{\text{ms}}, G)$  yields  $\Phi(M) \geq \tau$ . Since  $\Phi$  is increasing in  $K_2$ , replacing the unknown  $K_2$  by the certified upper bound  $47897/10000$  overestimates  $\Phi$ ; the strict failure  $\Phi(M_0) < \tau$  at the witness  $M_0 = 1292/1000$  therefore implies  $\Phi(R(f)) > \Phi(M_0)$ , which (by Lemma 2.5) forces  $R(f) > M_0$ .
- (4) *Strict failure.* By Proposition 5.1,  $\Phi(1292/1000) < \tau$ .
- (5) *Conclusion.* Strict monotonicity of  $\Phi$  on  $[1, \infty)$  (Lemma 2.5) together with (3) and (4) forces  $M > 1292/1000$ . Taking the infimum over  $f \in \mathcal{F}$  gives  $C_{1a} \geq 1292/1000 = 1.292$ .  $\square$

## 5.3. Lean Formalization

The analytic chain above is mechanized in Lean 4 across 30 modules totalling  $\approx 15.5$  thousand lines of code, which build cleanly under Mathlib [The20, dMU21] with zero `sorry` tactics. The formalization is organized in two layers. The *core* layer (13 modules, rooted at `lean/Sidon/MultiScale.lean`) factors the analytic reduction into `Sidon.{Defs, Bessel, FourierAux, TorusParseval, MVLemmas, MasterFromLemmas, BundleDefs, BundleEq1, BundleEq2Schwartz, BundleEq3Schwartz, BundleEq4, BilinearParseval, MultiScale}` and produces the conditional headline theorem `autoconvolution_ratio_ge_1292_1000`. A *constructor* layer (`Sidon.Constructor.{...}`) 17 further modules, all axiom-free) then assembles the unconditional headline `C1a_ge_1292_unconditional`. We describe the two headlines and their axiom budgets in turn.



**Conditional headline (bundle as hypothesis).** The theorem `autoconvolution_ratio_ge_1292_1000` carries an `ExtremiserPrimitives`  $f$  hypothesis record and reaches exactly *two* numerical user axioms in its dependency closure (alongside the three Lean core axioms `propext`, `Classical.choice`, `Quot.sound`):

- `K2_analytic_le_K2UpperQ`: the certified upper bound  $K_2(K_{\text{ms}}) \leq \text{K2UpperQ} = 47897/10000$  from Lemma 4.2. Here `K2_analytic` is the concrete real analytic functional  $\int_{\mathbb{R}} K_{\text{ms}}(x)^2 dx$  built from the explicit multi-scale kernel  $K_{\text{ms}}$  of Definition 3.1.
- `gain_analytic_ge_gainLowerQ`: the certified lower bound  $a \geq \text{gainLowerQ} = 20925/100000$  from Lemma 4.5. Here `gain_analytic` is the concrete real analytic functional  $(4/u) \cdot m_G^2/S_1$ , with  $m_G = \inf_{x \in [0, 1/4]} G(x)$  and  $S_1 = \sum_{j=1}^{200} a_j^2/\widetilde{K_{\text{ms}}}(j)$  both formed from the explicit cosine multiplier  $G_{\text{concrete}}(x) = \sum_{j=1}^{200} (\text{qpNumerators}[j-1]/10^8) \cos(2\pi jx/u)$  (the 200 rational coefficients embedded in `Sidon.MultiScale` as the integer list `qpNumerators`) and the closed-form Bessel-based periodic coefficients  $\widetilde{K_{\text{ms}}}(j) = \sum_i \lambda_i \text{besselJ0}(\pi j \delta_i/u)^2$  (using the power-series Bessel function from `Sidon.Bessel`).

Both axioms bind concrete real expressions in the Lean term language and are decidable in principle by any sufficient interval-arithmetic implementation; they play the role of MV 2010’s “Mathematica computed this value” citations [MV10], with the underlying numerical work performed here by the `flint.arb` certifier at 256-bit precision (in `delsarte_dual/grid_bound_alt_kernel/`). The three-scale master inequality at the analytic anchors is itself a Lean *theorem* (`MV_master_inequality_from_MV_lem` composed from the four MV-Lemma-3.1 atomic conclusions formalized in `Sidon.MVLemmas`); slack-monotonicity then lifts it to the rational anchors used in the strict-failure step.

**Unconditional headline (bundle constructed).** The constructor layer `Sidon.Constructor` mechanizes the construction of the `ExtremiserPrimitives`  $f$  record from raw admissibility, so that the bundle is no longer assumed but *built*. Its 17 modules are axiom-free and supply, for any admissible  $f$ : the  $L^1$  convolution Fourier identity, the membership  $K_{\text{ms}} \in L^2$  via the Young estimate  $L^{4/3} \star L^{4/3} \rightarrow L^2$ , a period- $u$  Poisson sampling lemma (generalizing `mathlib`’s period-1/continuous version), MO 2009 Lemma 2.1 in period- $u$  Parseval form, the period-1 Parseval identity for  $f \circ f$ , and the discrete Cauchy–Schwarz floor of (4). These are stitched into the bundle by `ExtremiserPrimitives.of_admissible`, and feeding the result into the conditional headline yields the unconditional theorem `C1a_ge_1292_unconditional`. It takes *only* the admissibility hypotheses —  $f \in L^1 \cap L^2$ ,  $\text{supp } f \subseteq (-1/4, 1/4)$ ,  $f \geq 0$ ,  $\int f = 1$  — and carries no hypothesis record. Its dependency closure reaches the same three Lean core axioms together with *four* numerical user axioms: the two above, plus

- `min_G_analytic_ge_minGLowerQ`: the certified bound  $m_G = \inf_{x \in [0, 1/4]} G_{\text{concrete}}(x) \geq 998/1000$  from Lemma 4.4 (a numerical input that the conditional headline instead absorbs through the bundle’s gain field); and
- `K_ms_fourier_lattice_pos_active`: the certified active-set positivity  $\widetilde{K_{\text{ms}}}(j) > 0$  for every integer  $j \in [1, 200]$ , which licenses the division by  $\widetilde{K_{\text{ms}}}(j)$  in  $S_1$  on the Lean side.

All four are certifier outputs of the same form — “`flint.arb` evaluated this functional” — decidable in principle by any sufficient interval-arithmetic implementation. Because the constructor layer is axiom-free, the unconditional headline is a strict strengthening of the conditional one: it reaches the same mathematical conclusion  $C_{1a} \geq 1292/1000$  with the analytic bundle now *proved* in Lean for every admissible  $f$  rather than *assumed*, at the cost only of the two additional decidable numerical facts.

**Sound definitions of  $f \circ f$  and  $K_{\text{arc}}$ .** The Lean formalization defines the convolutional autocorrelation as  $(f \circ f)(x) := \int_{\mathbb{R}} f(t) f(x+t) dt$  (matching the MV notation [MV10, §3]; this is `autocorr` in `Sidon.FourierAux`), and the arcsine kernel as the autoconvolution  $K_{\text{arc}}(\delta, \cdot) := \eta_\delta * \eta_\delta$  of the half-arcsine density  $\eta_\delta$  supported on  $(-\delta/2, \delta/2)$ . This matches the analytic definitions used throughout Sections 2–3 of the present paper and the Python certifier.

**Theorem statements.** The conditional headline theorem `autoconvolution_ratio_ge_1292_1000` carries a record `ExtremiserPrimitives f` whose fields are Lean restatements of the outputs of MO 2009 Lemmas 3.1–3.4 / MV 2010 Lemma 3.1 at the pair  $(f, K_{\text{ms}})$ ; those lemmas apply to  $K_{\text{ms}}$  directly (as a pdf supported in  $[-\delta_1, \delta_1]$  with  $\widetilde{K}_{\text{ms}}(j) \geq 0$  and  $K_{\text{ms}} \in L^2$ ), and the paper discharges the bundle fields by direct citation, exactly as MV 2010 cited MO 2009 at single arcsine. The unconditional headline `C1a_ge_1292_unconditional` removes even the named record: the constructor layer described above builds it from raw admissibility, so the field-by-field  $L^1 \cap L^2$  Plancherel and period- $u$  Parseval content (for which `mathlib` v4.29.1 exposes no single-call API) is supplied by the purpose-built `Sidon.Constructor` modules rather than assumed. The bundle’s  $m_G$  and  $S_G$  fields are reals constrained by the gain identity `gain_analytic = 2 m_G^2 / S_G` (field `gain_eq`); the bundle’s third inequality field `hEq3_ge` (replacing an earlier equality form `hEq3`) is Lean-derivable from finite- $J$  Parseval plus Bochner positivity  $\widetilde{K}_{\text{ms}}(j) \geq 0$  alone, via the supporting theorems `mv_eq3_ge` and `mv_eq3_ge_of_eq` in `Sidon.MVLemmas` (no period- $u$  Poisson summation required). A Schwartz-class specialisation that previously bypassed the `ExtremiserPrimitives` hypothesis was retired as vacuous: Paley–Wiener combined with Carlson’s theorem prevents any nontrivial Schwartz  $f$  from being compactly supported in  $(-1/4, 1/4)$ , so the Schwartz headline had no nontrivial witness.

**Rigor relative to Matolcsi–Vinuesa.** Taken together, the two headlines establish  $C_{1a} \geq 1.292$  to at least the standard of rigor of Matolcsi–Vinuesa [MV10], and in several respects beyond it. The analytic content is discharged both by direct citation to MO 2009 / MV 2010 applied to  $K_{\text{ms}}$  (as MV did at single arcsine) *and*, in the unconditional headline, mechanically constructed in Lean for every admissible  $f$ . The numerical content is the same *kind* of computer-checked input as MV’s Mathematica citations, but produced by `flint.arb` midpoint–radius interval arithmetic at 256-bit precision [Joh17] (rigorous guaranteed enclosures, not heuristic floating point) and independently corroborated in `mpmath`. The final assembly is carried out in exact rational arithmetic with a strictly positive closing margin: the master inequality’s true threshold is  $M_* = 1.29203$ , so the rational headline target 1.292 is a safe round-down, and the strict-failure margin  $\tau - \Phi(1292/1000) \geq 9.6 \times 10^{-5}$  of Proposition 5.1 is verified rationally. We retain the honest caveat that four computer-checked numerical axioms remain in the unconditional headline (two in the conditional one): the result is computer-assisted in the standard sense (cf. Flyspeck for Kepler [Joh17]), not a pure-kernel hand proof.

Six further statements are Lean *theorems*: the algebraic inversion `master_inequality_M_lower` (corresponding to Proposition 5.1, proved by case analysis on  $M \leq 1$  versus  $M > 1$  using `Real.sqrt` monotonicity) and the five slack-soundness theorems `K_two_upper_bound`, `k_one_lower_bound`, `S_one_upper_bound`, `min_G_lower_bound`, `gain_lower_bound`, each a one-line `norm_num` verification that the rational slack is on the correct side of the certifier-reported decimal recorded in Lemmas 4.1–4.5.

An explicit finiteness hypothesis  $\|f * f\|_{L^\infty} < \infty$  appears in the Lean statement because of the `ENNReal.toReal` encoding; this is harmless for the infimum. The support hypothesis is stated as `Function.support f ⊆ (-1/4, 1/4)`, i.e.  $f$  vanishes pointwise off  $(-1/4, 1/4)$ ; for any  $f \in \mathcal{F}$  in the

sense of Section 1.2, its representative obtained by zero-extending off the essential support satisfies the pointwise condition and has the same autoconvolution ratio, so the Lean bound transfers to the  $L^1$ -equivalence-class formulation of  $C_{1a}$  without loss.

## Acknowledgments

We thank Máté Matolcsi and Carlos Vinuesa, whose dual framework [MV10] is the foundation of this argument; and Alexander Cloninger and Stefan Steinerberger [CS17], whose announced bound of 1.28 motivated this line of work. The formalization rests on the Lean 4 theorem prover [dMU21] and the Mathlib library [The20]; the certifier rests on FLINT/Arb [Joh17]. We thank the authors and contributors of each.

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